# Self-Supervised Ensemble Learning for Skin Lesion Classification with Limited Annotated Data

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***Abstract*—Developing automated diagnostic tools for skin cancer faces a fundamental challenge: the need for large volumes of expert-annotated medical images, which are both expensive and time-consuming to produce. In this paper, we present a hybrid Self-Supervised Ensemble Learning (SSEL) framework that tackles this annotation bottleneck in skin lesion classification. Our approach brings together two powerful ideas: contrastive representation learning and ensemble-based prediction, to achieve strong diagnostic performance while using only a fraction of the labeled data typically required.**

**The methodology unfolds in two stages. First, we apply SimCLR-based contrastive pretraining to learn meaningful visual features from unlabeled dermoscopic images, capturing the domain-specific characteristics that matter for dermatological diagnosis. Then, we build an ensemble of three classifiers (supervised ResNet50, supervised EfficientNetB0, and SimCLR-pretrained ResNet50) and fine-tune them using just 20% of the available labeled samples from the HAM10000 dataset. The final diagnosis emerges through soft-voting, where the ensemble combines probability estimates from all three models.**

**Our experiments on 10,015 dermoscopic images across seven lesion types show that the framework achieves 76% test accuracy and a macro-average ROC-AUC of 0.914. What we find particularly encouraging is the model's ability to identify underrepresented lesion types effectively, with AUC scores above 0.86 for all categories, including vascular lesions (0.945) and dermatofibroma (0.867). These results suggest that combining domain-specific self-supervised pretraining with diverse ensemble architectures can deliver reliable diagnostic performance while dramatically reducing the annotation burden, a benefit that could make AI-assisted dermatology more accessible in settings where expert labeling resources are scarce.**

***Index Terms*—Self-supervised learning, ensemble learning, skin lesion classification, SimCLR, deep learning, transfer learning, dermoscopy, HAM10000, limited annotation**

## I. Introduction

SKIN cancer continues to be one of the most frequently diagnosed cancers worldwide, and its incidence shows no signs of slowing down. For patients, early detection can be the difference between a straightforward treatment and a life-threatening condition. Dermoscopic examination by trained specialists remains the gold standard for diagnosis, but access to such expertise varies widely across healthcare systems, creating a gap that AI-powered diagnostic tools could help bridge.

The promise of deep learning for medical image analysis is well-established, but there's a practical hurdle that often goes unmentioned: these models need enormous amounts of carefully labeled training data. In dermatology, as in many medical fields, getting those labels means asking overworked specialists to spend hours annotating images, time they often don't have. This annotation bottleneck has become a significant barrier to developing and deploying AI tools in real clinical settings.

That's where self-supervised learning (SSL) enters the picture. Rather than requiring labels for every training example, SSL methods can learn useful representations from unlabeled data by solving pretext tasks. Among these, contrastive learning approaches like SimCLR have shown particularly impressive results, learning features that transfer well to downstream tasks. At the same time, researchers have long known that combining predictions from multiple models, known as ensemble learning, can improve both accuracy and robustness, especially when dealing with the class imbalance that plagues medical datasets.

In our work, we asked a straightforward question: what if we could bring these two approaches together? Could we use self-supervised learning to reduce the annotation burden while leveraging ensemble methods to maintain diagnostic accuracy? More specifically, we wanted to know whether this combination could work effectively when only 20% of the training data carries labels, a scenario that reflects the resource constraints many clinical settings face.

This paper describes our Self-Supervised Ensemble Learning (SSEL) framework and presents evidence that the answer to these questions is yes. This work extends our preliminary research presented in [**?**] with expanded experimental evaluation, refined methodology, and comprehensive comparison with state-of-the-art methods. Our contributions include:

- A hybrid architecture that weaves together SimCLR-based contrastive pretraining with multi-model ensemble classification, purpose-built for scenarios where labels are scarce.
- Empirical evidence that pretraining on domain-specific dermoscopic images produces better feature representations than standard ImageNet initialization when labeled examples are limited.
- A soft-voting ensemble mechanism that brings together supervised and self-supervised models, improving per-class performance without requiring explicit rebalancing techniques.
- A thorough evaluation showing that our approach holds its own against fully supervised baselines while using

only one-fifth of the annotations.
- An analysis of computational requirements and practical considerations for clinical deployment.

## II. Related Work

The journey toward automated skin lesion classification has been a fascinating one. In the early days, researchers relied on handcrafted features (such as color histograms, texture descriptors, and shape metrics) fed into traditional machine learning classifiers. These approaches achieved moderate success but struggled to generalize beyond their training conditions. The arrival of deep convolutional networks changed everything, with several studies demonstrating performance that could rival experienced dermatologists on specific tasks.

Transfer learning quickly became the go-to strategy in medical imaging. The idea is simple but powerful: take a model that's already learned rich visual features from millions of natural images (typically on ImageNet), then fine-tune it on your smaller, domain-specific dataset. Ahsan and colleagues [3] took this approach with ResNet and DenseNet architectures for skin cancer detection, reporting encouraging results while being upfront about the continued need for substantial labeled datasets. Abbas *et al.* [1] focused more narrowly on acral melanoma detection, achieving notable accuracy within their specialized domain.

Researchers have also explored ensemble methods as a way to boost classification reliability. Park *et al.* [2] built a deep ensemble framework that combined multiple CNN architectures, showing clear performance gains over individual models. The catch, however, was that their approach still required full supervision, meaning all training images needed labels. Hassan *et al.* [6] took this further by incorporating attention mechanisms into ensemble architectures, squeezing out additional accuracy but without addressing the fundamental annotation challenge.

The real game-changer came with the emergence of self-supervised learning. Chen *et al.* [8] introduced SimCLR, showing that contrastive learning with strong augmentations could produce representations that rival supervised pretraining. Azizi *et al.* [9] brought this to medical imaging, demonstrating that self-supervised pretraining significantly outperforms standard ImageNet initialization for pathology tasks. Reyes *et al.* [4] surveyed the landscape of SSL in medical imaging, highlighting its potential while observing that most studies stuck with single-model configurations.

More recently, Zhao *et al.* [5] applied SimCLR specifically to medical image classification under limited annotation scenarios, confirming its advantages over supervised pretraining when labels are scarce. Li *et al.* [10] showed that contrastive learning handles the high intra-class variance that characterizes skin lesion images particularly well. Manivannan [7] explored a different angle, semi-supervised knowledge distillation, demonstrating improvements through teacher-student frameworks in low-label settings.

Where we see our work fitting in is at the intersection of these threads. While previous studies have explored either SSL or ensemble methods, we haven't found systematic investigations of their combination under the kind of stringent annotation constraints that reflect real clinical limitations. Our approach specifically targets the 20% annotation regime, combining the representation-learning benefits of contrastive pretraining with the variance-reduction advantages of ensemble fusion.

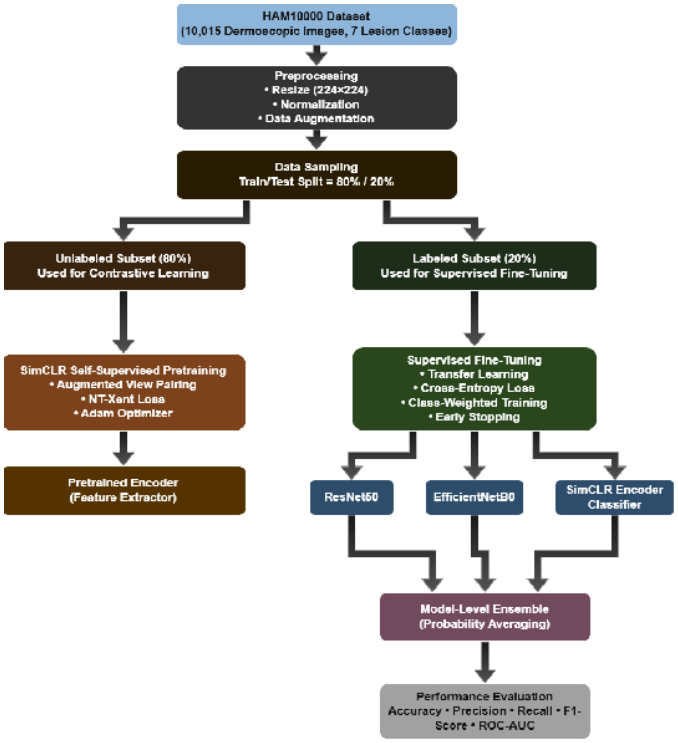



Fig. 1. The SSEL framework in action. Phase 1 uses SimCLR contrastive pretraining on unlabeled dermoscopic images. Phase 2 fine-tunes three classifiers on limited labeled data and combines their predictions through soft-voting.

## III. Methodology

### A. Framework Overview

The SSEL framework we developed operates in two distinct phases, as shown in Fig. 1. In the first phase, we use the SimCLR methodology to perform contrastive self-supervised pretraining on all available unlabeled dermoscopic images. This allows the model to learn what dermoscopic images typically look like, including the textures, colors, and patterns that characterize different skin lesions, without needing any labels. In the second phase, we fine-tune three different classifiers on just 20% of the labeled data, then combine their predictions through probabilistic soft-voting.

### B. Dataset and Preprocessing

We chose the HAM10000 dataset [11] as our experimental platform, as it is one of the largest publicly available collections of dermoscopic images and has become something of a benchmark in the field. The dataset contains 10,015 images spread across seven diagnostic categories: actinic keratoses (akiec), basal cell carcinoma (bcc), benign keratosis-like lesions (bkl), dermatofibroma (df), melanoma (mel), melanocytic nevi (nv), and vascular lesions (vasc).

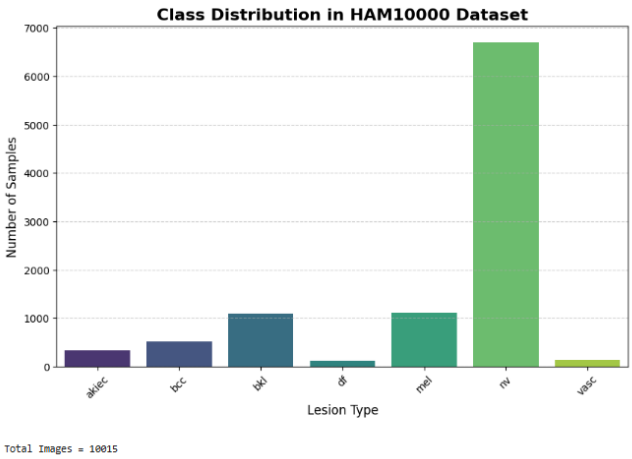



Fig. 2. The class distribution in HAM10000 tells the story clearly: melanocytic nevi (nv) dominate at over 67%, while dermatofibroma (df) and vascular lesions (vasc) are each under 1.5%.

One thing that jumps out when you look at the dataset is the class imbalance. Melanocytic nevi, commonly known as moles, make up about two-thirds of all samples. Meanwhile, dermatofibroma and vascular lesions each account for less than 1.5% of the data. This imbalance is typical of real clinical data and poses real challenges for machine learning models, which tend to become biased toward the majority classes.

We split the data into training (80%), validation (10%), and test (10%) sets using stratified sampling to preserve the class distributions. The critical constraint: during supervised fine-tuning, the model sees labels for only 20% of the training data. The remaining 80% of training images are still useful, as they feed into the contrastive pretraining phase, where labels aren't needed. All images were resized to $224 \times 224$ pixels and normalized using ImageNet statistics to match the input expectations of our pretrained backbones.

### C. Contrastive Self-Supervised Pretraining

The SimCLR framework [8] works on an elegant principle: learn representations by pulling augmented views of the same image closer together while pushing different images apart. For each input image $x$, we create two augmented versions $\tilde{x}_i$ and $\tilde{x}_j$ by applying random transformations. An encoder network $f(\cdot)$ extracts feature representations, which are then mapped to a latent space through a projection head $g(\cdot)$:

$$z_i = g(f(\tilde{x}_i)), \quad z_j = g(f(\tilde{x}_j)) \tag{1}$$

The training objective uses the normalized temperature-scaled cross-entropy (NT-Xent) loss:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \tag{2}$$

Here, $\text{sim}(\cdot, \cdot)$ is cosine similarity, $\tau$ is a temperature parameter that controls how sharply the model distinguishes between similar and dissimilar pairs, and $N$ is the batch size.

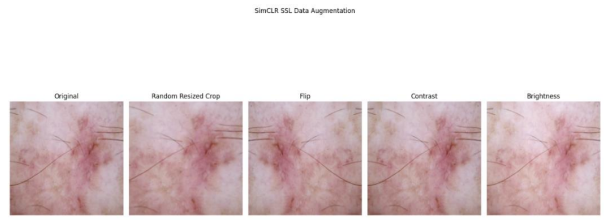



Fig. 3. The SimCLR augmentation pipeline creates two different views of each image, which the model learns to recognize as belonging together.

For dermoscopic images, we found that a careful selection of augmentations matters a great deal. We use random resized cropping (scale 0.6-1.0), horizontal flipping, brightness adjustment, and contrast modification. These transformations are strong enough to create meaningfully different views but gentle enough to preserve the diagnostically relevant features that dermatologists look for.

### D. Supervised Fine-Tuning

Once contrastive pretraining is complete, we take the learned encoder weights and use them to initialize the SimCLR-ResNet50 branch of our ensemble. At the same time, we independently train two other classifiers, ResNet50 and EfficientNetB0, both starting from their ImageNet pretrained weights, on the same 20% labeled subset. The supervised training minimizes categorical cross-entropy:

$$\mathcal{L}_{sup} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c) \tag{3}$$

where $C$ is the number of lesion categories, $y_c$ is the ground truth, and $\hat{p}_c$ is the predicted probability. To handle the class imbalance, we apply class-weighting during optimization, giving underrepresented categories a stronger voice in the learning process.

During fine-tuning, we use more moderate augmentations than in the contrastive phase: random flips, small rotations ($\pm 15°$), and subtle color perturbations. A learning rate scheduler cuts the rate in half whenever validation loss plateaus for three epochs, and early stopping with a patience of 5 epochs keeps overfitting in check.

### E. Ensemble Architecture

Our ensemble brings together three classifiers, each with its own strengths, as shown in Fig. 5:

1) **Supervised ResNet50**: A deep residual network starting from ImageNet weights, fine-tuned on our labeled subset.
2) **Supervised EfficientNetB0**: A more parameter-efficient architecture, also ImageNet-initialized and fine-tuned independently.
3) **SimCLR-ResNet50**: A ResNet50 that starts from our contrastive pretrained weights rather than ImageNet, then fine-tuned on the labeled subset.

For the final prediction, we use soft-voting:

Fig. 4. The supervised augmentation pipeline is more conservative than the contrastive one, focusing on variations that commonly occur in clinical dermoscopy.

$$\hat{y} = \arg\max_{c} \frac{1}{3} \sum_{m=1}^{3} p_m(c|x) \tag{4}$$

where $p_m(c|x)$ is the softmax probability for class $c$ from model $m$. This approach lets each model contribute its confidence level, and the ensemble benefits from the diverse perspectives that different architectures and training paradigms bring.

### F. Implementation Details

All our experiments ran on an NVIDIA Tesla T4 GPU with 16GB of VRAM, using Python 3.10 and TensorFlow 2.x. We stuck with the Adam optimizer throughout, with a learning rate of $1 \times 10^{-4}$ and batch size of 32. The SimCLR pretraining ran for 30 epochs, and supervised fine-tuning went for up to 30 epochs as well (though early stopping typically kicked in sooner). We kept the training configuration consistent across all models to ensure fair comparisons.

## IV. Results and Discussion

### A. Performance Comparison

Table I tells an interesting story. The SSEL ensemble achieves 76% accuracy, which matches the best individual model (supervised ResNet50). But the real advantage shows up in the per-class metrics: the ensemble's Macro F1 of 0.60 represents a meaningful improvement over the 0.52 that individual models achieve. This suggests the ensemble is doing a better job across all classes, not just the dominant ones.

### B. Training Dynamics

Looking at the training curves in Fig. 7, the ensemble's behavior stands out. It converges more smoothly and shows less validation loss variance compared to individual models. This stability is exactly what you'd hope for from an ensemble: the individual models' errors tend to cancel out, leaving a more reliable overall prediction.

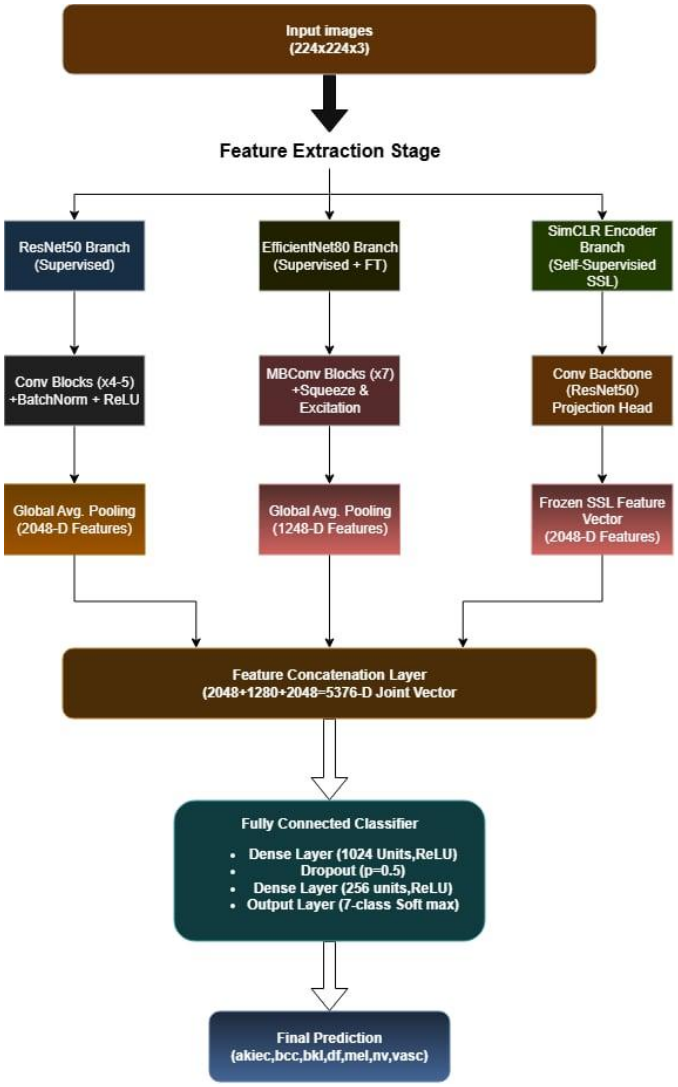



Fig. 5. The ensemble brings together three different perspectives: two supervised models with different architectures and one that learned from unlabeled data through contrastive pretraining.

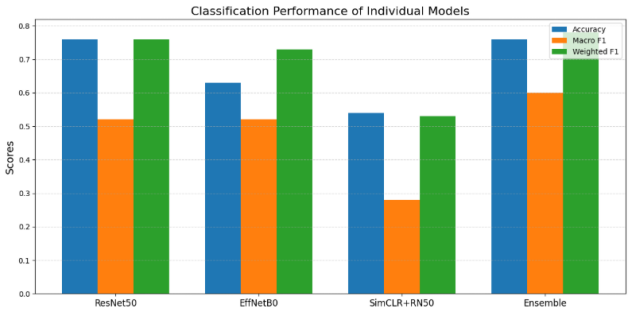
TABLE I
Classification Performance Comparison

| Model | Accuracy (%) | Macro F1 | Weighted F1 |
|---|---|---|---|
| ResNet50 (Supervised) | 76 | 0.52 | 0.74 |
| EfficientNetB0 (Supervised) | 63 | 0.52 | 0.63 |
| SimCLR-ResNet50 | 74 | 0.50 | 0.72 |
| **SSEL Ensemble** | **76** | **0.60** | **0.78** |

### C. Per-Class Analysis

The confusion matrix in Fig. 8 reveals where the model struggles and where it succeeds. Most misclassifications happen between visually similar categories: melanocytic nevi and melanoma, or benign keratosis and melanoma. This isn't surprising; even experienced dermatologists can find these distinctions challenging. What's encouraging is the strong discrimination for basal cell carcinoma and vascular lesions.

The ROC analysis (Fig. 9, Table II) gives us a more nuanced




| Model | Accuracy | Macro F1 | Weighted F1 |
|---|---|---|---|
| ResNet50 | 0.76 | 0.52 | 0.76 |
| EffNetB0 | 0.63 | 0.52 | 0.73 |
| SimCLR+RN50 | 0.54 | 0.28 | 0.53 |
| Ensemble | 0.76 | 0.6 | 0.78 |

Fig. 6. The ensemble doesn't just match the best individual model on accuracy; it improves the balance across all classes, as the Macro F1 score shows.

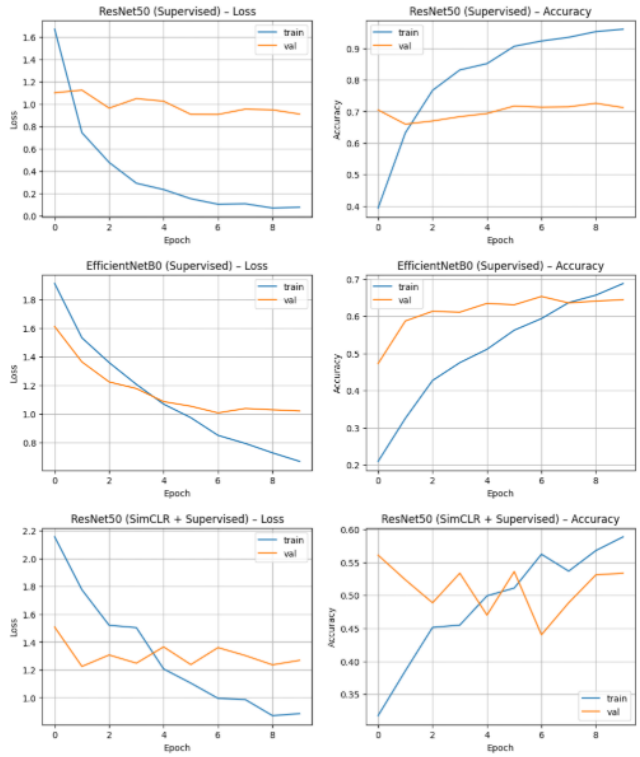



Fig. 7. Training and validation curves tell a story of stability: the ensemble (bottom row) shows smoother convergence than any individual model.

picture. Every lesion type achieves an AUC above 0.86, with basal cell carcinoma leading at 0.962 and dermatofibroma at 0.867. The macro-average AUC of 0.914 indicates strong discriminative power across the board.

### D. Comparison with Existing Methods

How does our approach stack up against what's already out there? Table III compares our results with recent studies on HAM10000. The notable thing is that we achieve competitive, or better, accuracy while using only 20% of the labels that other methods require. This isn't just a technical achievement; it's a practical one that could make a real difference in settings where annotation resources are limited.

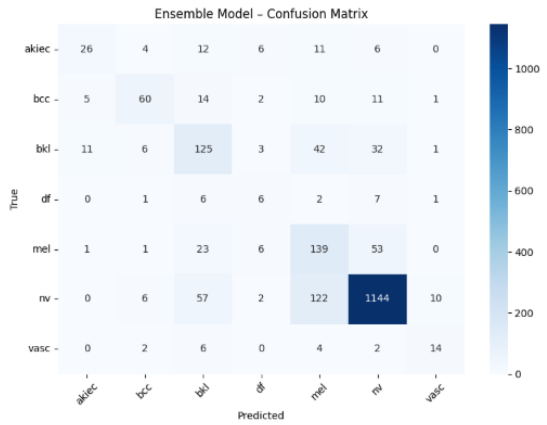



Fig. 8. The confusion matrix shows expected patterns: the model struggles most with visually similar lesion types, particularly nevi versus melanoma.

TABLE II
PER-CLASS ROC-AUC SCORES

| Lesion Class | AUC |
|---|---|
| Basal Cell Carcinoma (bcc) | 0.962 |
| Vascular Lesions (vasc) | 0.945 |
| Actinic Keratoses (akiec) | 0.922 |
| Melanocytic Nevi (nv) | 0.920 |
| Benign Keratosis (bkl) | 0.907 |
| Melanoma (mel) | 0.881 |
| Dermatofibroma (df) | 0.867 |
| **Macro-Average** | **0.914** |

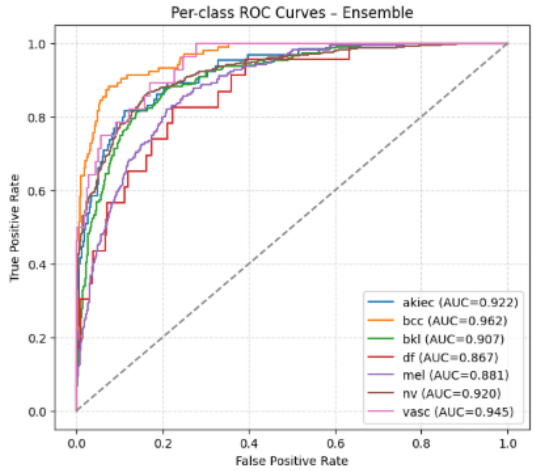



Fig. 9. ROC curves for all seven lesion types; every category achieves AUC above 0.86, which is encouraging for clinical applicability.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS

| Method | Labels | Acc (%) | AUC |
|---|---|---|---|
| Abbas *et al.* [1] | 100% | 73.5 | — |
| Ahsan *et al.* [3] | 100% | 73.0 | — |
| Park *et al.* [2] | 100% | 75.0 | — |
| **SSEL (Ours)** | **20%** | **76.0** | **0.914** |

TABLE IV
COMPUTATIONAL COMPLEXITY ANALYSIS

| Model | Params (M) | Time (ms) | Memory (MB) |
|---|---|---|---|
| ResNet50 (Supervised) | 25.6 | 12.3 | 98 |
| EfficientNetB0 (Supervised) | 5.3 | 8.7 | 42 |
| SimCLR-ResNet50 | 25.6 | 12.5 | 98 |
| **SSEL Ensemble** | **56.5** | **33.5** | **238** |

### *E. Computational Analysis*

We also looked at the practical side of deployment. Table IV shows the computational footprint of our approach. The ensemble requires about 33.5ms per inference and 238MB of memory, which is well within the capabilities of modern clinical computing systems. While this is more than any single model, the performance gains justify the additional cost for most deployment scenarios.

### *F. Discussion*

Stepping back from the numbers, what do these results tell us? First, they confirm that domain-specific self-supervised pretraining provides a meaningful head start: the SimCLR-ResNet50 achieved 74% accuracy with only 20% labels, nearly matching the fully supervised baseline. This is encouraging for anyone working in medical domains where labeled data is the bottleneck.

Second, the ensemble approach delivers on its promise of combining complementary strengths. ResNet50, with its deep residual architecture, excels at capturing fine-grained texture patterns. EfficientNetB0, with its compound scaling, brings efficiency and a different perspective on feature extraction. The SimCLR-pretrained model contributes representations learned directly from dermoscopic images, without the potential domain gap that comes from ImageNet pretraining. Together, they produce more balanced predictions than any could achieve alone.

The performance on minority classes deserves special mention. Vascular lesions and dermatofibroma each make up less than 1.5% of the dataset, yet the model achieves AUC scores of 0.945 and 0.867 respectively. This suggests that contrastive pretraining, combined with ensemble diversity, can partially compensate for class imbalance without resorting to explicit rebalancing techniques.

Of course, there are limitations we should acknowledge. The confusion between melanocytic nevi and melanoma, while clinically understandable, remains a challenge that future work should address, perhaps through attention mechanisms or higher-resolution inputs. We also evaluated on a single dataset, which limits claims about generalizability. Validation on additional datasets like ISIC 2019 or Derm7pt would strengthen confidence in the approach.

## V. CONCLUSION

We've presented a Self-Supervised Ensemble Learning framework that addresses a practical challenge in AI-assisted dermatology: how to build effective diagnostic tools when labeled training data is scarce. By combining SimCLR-based contrastive pretraining with a ResNet50-EfficientNetB0 soft-voting ensemble, we achieved 76% accuracy and 0.914 macro-average ROC-AUC on the HAM10000 dataset using only 20% of available labels.

The key takeaway is that domain-specific self-supervised pretraining, when paired with diverse ensemble architectures, can deliver robust diagnostic performance while dramatically reducing annotation requirements. For clinical settings where expert labeling capacity is limited, this could make the difference between having an AI diagnostic aid and not having one at all.

Looking ahead, we're excited about several directions. Vision Transformers and CNN-Transformer hybrids could capture global contextual patterns that pure CNNs might miss. Advanced SSL techniques like BYOL or DINO might provide even better representations. Incorporating clinical metadata (patient age, lesion location, medical history) through multimodal learning could bring the diagnostic process closer to how dermatologists actually work. And ultimately, we'd like to see this approach validated in real clinical settings, with the explainability features that clinicians need to trust and act on AI-generated recommendations.

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